# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The number of factors of 48 is: 1

**(a)** 12 **(b)** 4 **(c)** 10 **(d)** 5

**2.** The HCF of 36 and 48 is: 1

**(a)** 12 **(b)** 144 **(c)** 36 **(d)** 24

**3.** The LCM of 12 and 18 is: 1

**(a)** 18 **(b)** 72 **(c)** 36 **(d)** 216

**4.** The smallest digit to put in the blank to make 452_ divisible by 3 is: 1

**(a)** 5 **(b)** 3 **(c)** 1 **(d)** 2

**5.** The largest 3-digit multiple of 7 is: 1

**(a)** 999 **(b)** 994 **(c)** 987 **(d)** 980

**6.** How many prime numbers are there between 20 and 60? 1

**(a)** 9 **(b)** 10 **(c)** 11 **(d)** 8

**7.** The sum of all distinct prime factors of 210 is: 1

**(a)** 17 **(b)** 20 **(c)** 14 **(d)** 210

**8.** The number of factors of 54 is: 1

**(a)** 3 **(b)** 4 **(c)** 8 **(d)** 10

**9.** The HCF of 45 and 75 is: 1

**(a)** 30 **(b)** 225 **(c)** 15 **(d)** 45

**10.** The LCM of 8 and 20 is: 1

**(a)** 40 **(b)** 20 **(c)** 80 **(d)** 160

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Find the HCF and LCM of 24 and 36 using prime factorisation. 2

**12.** Find the smallest number that must be added to 1234 to make it divisible by 9. 2

**13.** Find the missing digit so that 456_ is divisible by 9. 2

**14.** Find the HCF and LCM of 30 and 45 using prime factorisation. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Find the prime factorisation of 180 and the number of its factors. 3

**16.** Two bells ring every 12 minutes and every 18 minutes. They ring together at 9 am. When do they next ring together? 3

**17.** Two ropes of length 48 m and 72 m are cut into pieces of equal length, as long as possible, with nothing left over. Find the length of each piece and the total number of pieces. 3

OR

Find the prime factorisation of 200 and the number of its factors. 3

**18.** Two bells ring every 15 minutes and every 20 minutes. They ring together at 9 am. When do they next ring together? 3

OR

Two ropes of length 60 m and 84 m are cut into pieces of equal length, as long as possible, with nothing left over. Find the length of each piece and the total number of pieces. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A room is 450 cm long and 360 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Find the smallest number that leaves a remainder 1 when divided by 4, 5 and 6, and the next such number. 5

**20.** A room is 540 cm long and 420 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Find the smallest number that leaves a remainder 1 when divided by 3, 4 and 8, and the next such number. 5

## END OF QUESTION PAPER — Total: 40 Marks

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